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Resume Analyzer based on MapReduce and Machine Learning

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***Abstract*— The AI Resume Analyzer is a groundbreaking AI-driven platform drafted to give job seekers an important array of tools and insights. At its core, this design offers a sophisticated resume analysis system that delivers personalized feedback, streamlining the process of tailoring resumes for specific job purposes and companies. This research paper presents an approach to analyze resumes by integrating MapReduce for efficient preprocessing and Feature Engineering, in combination with a machine learning model (K-Nearest Neighbors (KNN) Classifier) for job predictions. In this paper, resume dataset is being described briefly which contain the resume information related to several job roles, after this there is discussion about how map reduce can be used for preprocessing related task and parallel processing, followed by comparative analysis of the MapReduce on different subset of data. Finally, there is the discussion about how KNN is being used to make job predictions and analysis of the results obtained.**

Keywords*— *Machine Learning, MapReduce, Natural Language Processing, KNN, AI resume analyzer, Big Data.

I. INTRODUCTION

In the modern recruitment process the flow of digital resumes has increased, creating a need for organized techniques to process and extract valuable insights from resumes. A mixture of MapReduce- a parallel computing model- and AL driven NLP algorithms gives a promising solution to this need for efficient resume analysis [1]. The MapReduce model, adopted in systems such as Apache Hadoop, enables parallel computing of large chunks of data across clusters of computers [2]. It provides a very good framework for parallelizing tasks, making it perfect for handling the huge volume and type of resume data from various professional domains.

Similarly, machine learning algorithms like NLP have transformed language understanding and assessment. Algorithms uses machines to simplify and derive meaning from human language [3]. Based on the idea of resume analysis, NLP techniques help in extracting meaningful data such as qualifications, experiences, skills, interests, and goal which are important for job prediction and candidate assessment.

This research paper is made with the strengths of both MapReduce and AI driven NLP techniques. The aim is to develop a Resume Analyzer system that accurately processes diverse resumes using MapReduce for key word extraction and then uses NLP algorithms to predict jobs and compatibility based on the information extracted from resume. By mixing these advanced technologies, this paper aims to offer an innovative technique for resume analysis , increasing the speed and accuracy of candidate assessment and job prediction in the recruitment process.

II. LITERATURE SURVEY

There are several methods that can be used for processing datasets and resume analysis. Researchers have explored several methods for parallel and distributed processing. One study by Dean et al. [4] showed that how MapReduce programming model can be used for parallel processing of large datasets and explained the internal working of MapReduce architecture by working on 3,288 TB of data. Tomi A. [5] worked on a MapReduce processing model and showed that it can process terabytes of data on thousands of machines, reducing code size from 3800 to 700 lines and enabling GPU integration for much more computational speed. Lee et al. [6] proposed the parallel data processing MapReduce survey. They researched various technical aspects regarding MapReduce including its pros and cons and suggest other classification for MapReduce improvements. Khezr et al. [7] used MapReduce for the optimization algorithms. The paper presents some optimization algorithms like PSO, cuckoo search, genetic using in MapReduce applications. In [8] explained the widely used open-source runtime systems and shortcomings of original MapReduce implementation. They also have performed surveys on several optimization approaches. Kalavri et al. [9] provides a review of MapReduce with Hadoop architecture. This survey revolves around Hadoop and it underlying MapReduce framework. Andrew et al. [10] in their paper “MapReduce for Machine learning on multicore” worked on linear speed up with large number of processors on various Machine Learning algorithm. Lakshmi et al. [11] proposed an algorithmic model using Hadoop architecture for parallel implementation in single core CPUs. They found variation in time using serial and parallel computing of Machine learning techniques. Ghoting et al. [12] in their paper introduces NIMBLE, toolkit for implementing parallel data mining and machine learning algorithms on MapReduce. Woldemariam et al. [13] discussed syntactic parser and MapReduce framework running of several clusters of machines on cloud. Alamelu et al. [14] in their paper discussed how NLP algorithms can be used for text parsing and filtration of words from Resumes. Rakhi et al. [15] discussed about how NLP algorithm can be used for parsing and training machine learning model for Resume analysis. Poovizhi et al. [16] talk about Linear support vector machine, random field matching and NLP techniques to create automatic resume scraping system..

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III. METHODS AND TECHNOLOGY USED

A. Dataset

The dataset used in this work contains a wide range of publicly obtainable resumes that are collected from various online websites. The dataset contains a resume from various professional domains including resumes related to Information technology, business development, teaching, sales, etc. [18]. In this work, a large range of resumes were collected for each domain. Total number of resume were around 7000. The selection of these vide range ensure diversity in experience, educational background, and skills. The resumes collected were in different file format like pdf, txt, rtf etc. and for maintaining the similar formatting of data, the pre-processing of resumes was performed.

Resumes were collected from reliable online sources like Kaggle [17], job boards and career portals like JobSpider [18] where people publicly share their work experience and skill when applying for jobs. We made sure to follow privacy rules and removed any personal information from these resumes.

By gathering this large mix of resumes, we plan to create a dataset that contain many kinds of jobs and skills. This dataset will help us pre-process, build and test our Resume Analyzer system, and will make sure that it works well for various job types.

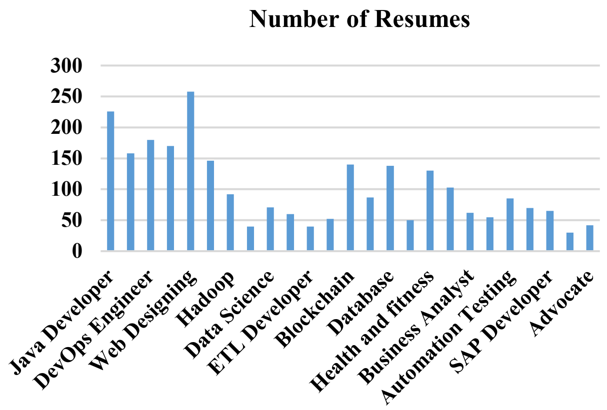

Fig. 1. Graph Showing Diversity of Dataset

B. Data Preprocessing

Data preprocessing is the first step done during solving a machine learning problem. To simplify the path of model development, preprocessing phase was done to make data consistent which was collected from resumes, which were also in different file formats. Our main aim at this stage was to convert various formats of resumes into some appropriate form, which were then used for further analysis. In this step we mainly discard the elements such as icons, special symbols, unnecessary data and inappropriate characters from the resumes.

Using text extraction and parsing techniques, we processed handled each resume and only retained the useful and textual information. Then we combined this data and created a single dataset in the CSV format. This CSV file has the information such as resume id, job position, and earlier pre-processed content from the resumes. The CSV file that was generated at this stage acted as the base dataset for the next stages of the model development. This consistent format supports simplifying the MapReduce techniques for feature engineering.

C. MapReduce Implementation

The MapReduce is a parallel programming model in which process large datasets by dividing bigger tasks into several nodes. It works by dividing the complex task into a map and reduces phase. Therefore, it helps in data processing in distributed systems like Hadoop that is scalable and fault tolerant [19]. These are the steps involved in the MapReduce Implementation:

1. CSV dataset file was uploaded to HDFS file system.
2. Map phase: For getting important information and removing unnecessary words.
3. Shuffle and sort phase: To sort the data and group them together based on job role.
4. Reduce phase: All the similar resume details related to specific job role are combined.

Role of Each MapReduce component

- Mapper Function: The dataset gets split into smaller parts in the Mapper phase. It extracts key information from resumes, like technical skills and important details, at the same time also cleaning out unnecessary words and elements.
- Shuffle and Sort Phase: Following the Mapper phase, data gets sorted and then organized based on the job roles to prepare for the next step, the Reduce phase.
- Reduce Function: In the final step of MapReduce, data related to jobs which have similar roles were merged. We took care to remove duplicate information across similar resumes to keep the data clear and concise.

D. Natural Language Processing implementation

NLP combines computational linguistics—rule-based modeling of human language—with statistical, machine learning, and deep learning models. Together, these technologies enable computers to process human language in the form of text or voice data and to 'understand' its full meaning, complete with the speaker or writer's intent and sentiment [21].

In the proposed work, to categorize resumes accurately and for using the NLP techniques we used K-Nearest Neighbors (KNN) classifier algorithm [23].

KNN is a widely used classification technique. It is commonly used for its easy interpretation and less calculation time. The choice of parameter k is very crucial in this algorithm. The training error rate and the validation error rate are two parameters which need to be accessed on different k values. The domain of text classification using KNN algorithm is very wide [22].

The dataset that has been received from MapReduce after pre-processing is split into training and testing sets. Afterwards, the KNN model is fitted or trained on the training set and further its performance is evaluated on the test set. Once the hyperparameter of the model is fine-tuned, the model is ready to classify and make predictions. The details of the implementation are given in the next section.

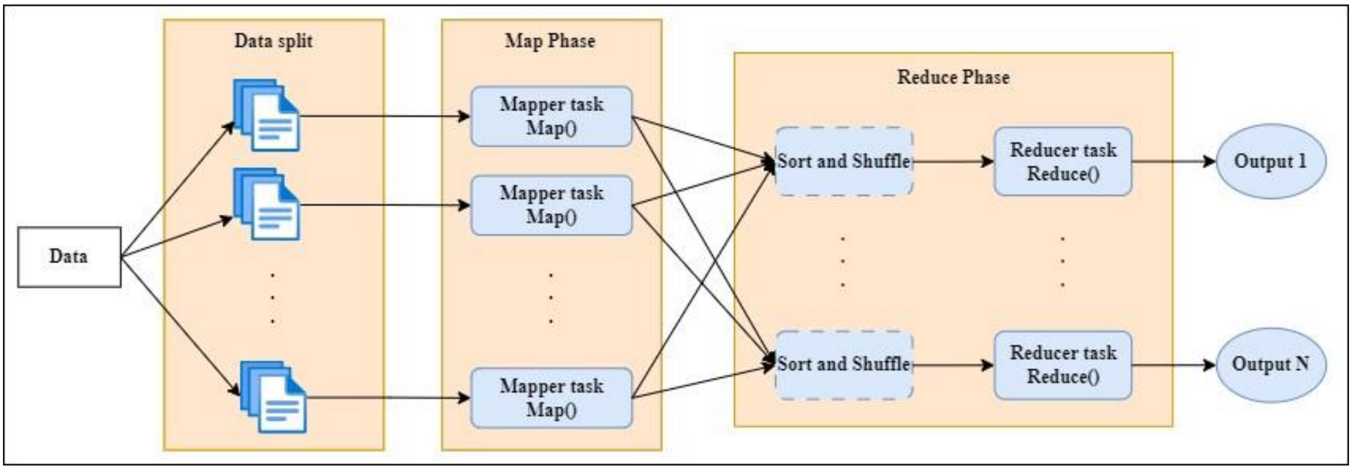

Fig. 2. MapReduce Architecture [20]

E. Proposed Methology

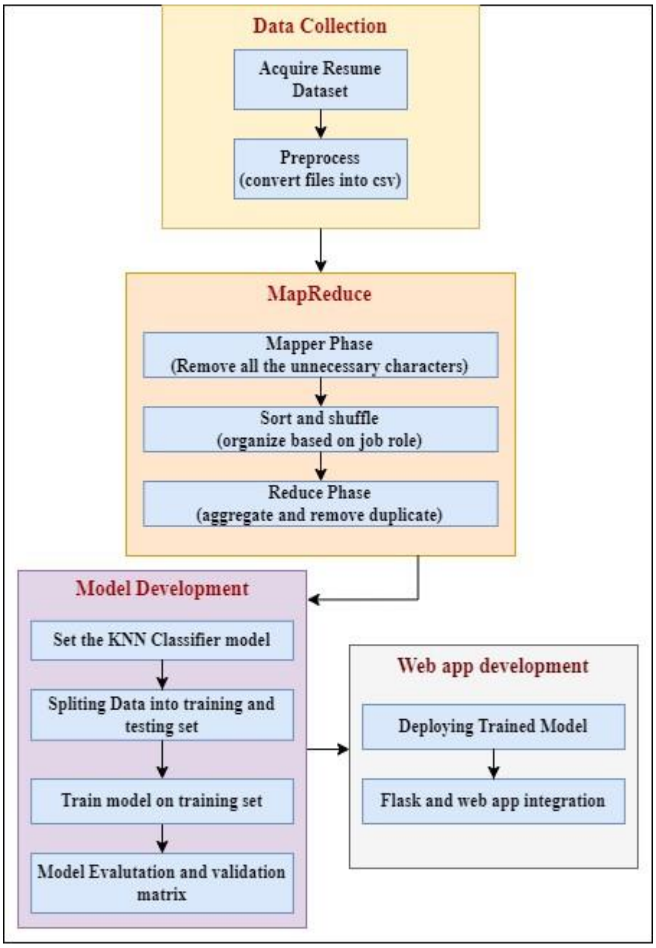

Fig. 3. Proposed Model

Following are the steps involved in proposed methodology:

1. Collected 7000 large variety of resumes for creating dataset.
2. Pre-process the dataset using MapReduce Framework. Different resume file formats were converted and data from all the text extracted were stored in a csv file.
3. Data presented into csv files were vectorized and converted into vectors that can be used for training models.
4. Divide the dataset into training set 80% and testing 20% set.
5. Import KNeighborsClassifier from scikit-learn package and train the model on the training set for text categorization.
6. Evaluate the model on test dataset based on accuracy, precision, recall and F1-score.

IV. RESULTS AND ANALYSIS

Under this section, first, there is a discussion about performance evaluation of MapReduce Processing on varying resume datasets, followed by the discussion about the results obtained for KNN classifier. The performance of the KNN model has been evaluated on four standard metrics: Accuracy, Precision, Recall and F1-score [24]. The confusion matrix of KNN is also given to justify the performance.

A. Performance Evaluation of MapReduce Processing on Varying Resume Datasets:

In the proposed study, MapReduce has been run on different amounts of data to check its effectiveness in processing data. The dataset has been divided into subsets with 100, 500, 1000, 2500, 3500 and 7000 resumes respectively. For each set, we have run MapReduce jobs on Hadoop and recorded how long it has been taken. We also considered CPU usage time, the amount of memory consumed, and number of mappers and reducers which have been involved during the processing.

These recorded times can help us in understanding how MapReduce performance speed changes with the changes in the amount of data provided. Also, by considering the data regarding resource consumption, important information about distributed environment can be obtained.

Table I and Table II show the number of mappers and reducers and time taken by each mapper and reducer to process the subsets of data and CPU usage. These observations give a breakdown of the processing aspect.

On the other hand, figure 2 provides a comparison of total time taken to process the various subsets within the dataset.

From Table I, the efficiency of MapReduce processing can be deduced. After observing the mapper and reducer time on different subset sizes, it becomes clear that when the amount of data increased, there is change in the time for processing. Also, the number of mappers and reducers affects the processing time on the dataset. For example, with 7000 resumes and with 4 mapper and reducers each, the mapper and reducer time decreased greatly in comparison to smaller set with 1 mapper and reducer.

Also, Table II shows the total time and CPU utilization during the MapReduce processing across different sizes of data. The total time has the same effect as of the Mapper and reducer time i.e., total time also decreased with the increase of dataset and increase of number of Mapper and reducer tasks. CPU time spent also aligns with the overall efficiency during the processing. Thus, from Table I, Table II, and Figure 4, we can confirm that MapReduce works better when a large amount of data is being given as input. By increasing the number of mappers and reducing tasks, processing time of MapReduce can significantly be increased.

TABLE I. Mapper and Reducer Time Elapsed

Subset size/ Observation	Mapper Time (in ms)	Reducer Time (in ms)	No. of Mappers	No. of Reducers
100 Resumes	5395	3589	1	1
500 Resumes	5401	3785	1	1
1000 Resumes	5273	3968	1	1
2500 Resumes	4808	3846	2	1
3500 Resumes	3531	3592	4	2
7000 Resumes	2476	2034	4	4

TABLE II.

Total MapReduce execution time and CPU time

Subset size/ Observation	Total Time (in ms)	CPU time spent (in ms)
100 Resumes	8984	3058
500 Resumes	9186	3214
1000 Resumes	9241	3385
2500 Resumes	8654	2891
3500 Resumes	7123	2656
7000 Resumes	4510	2142

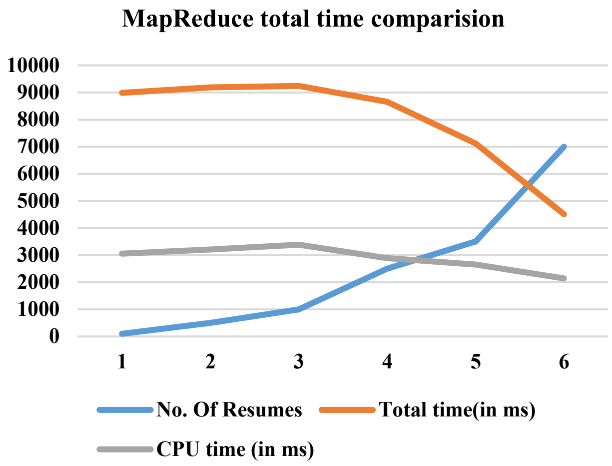

Fig. 4. MapReduce time comparison for different subsets of data

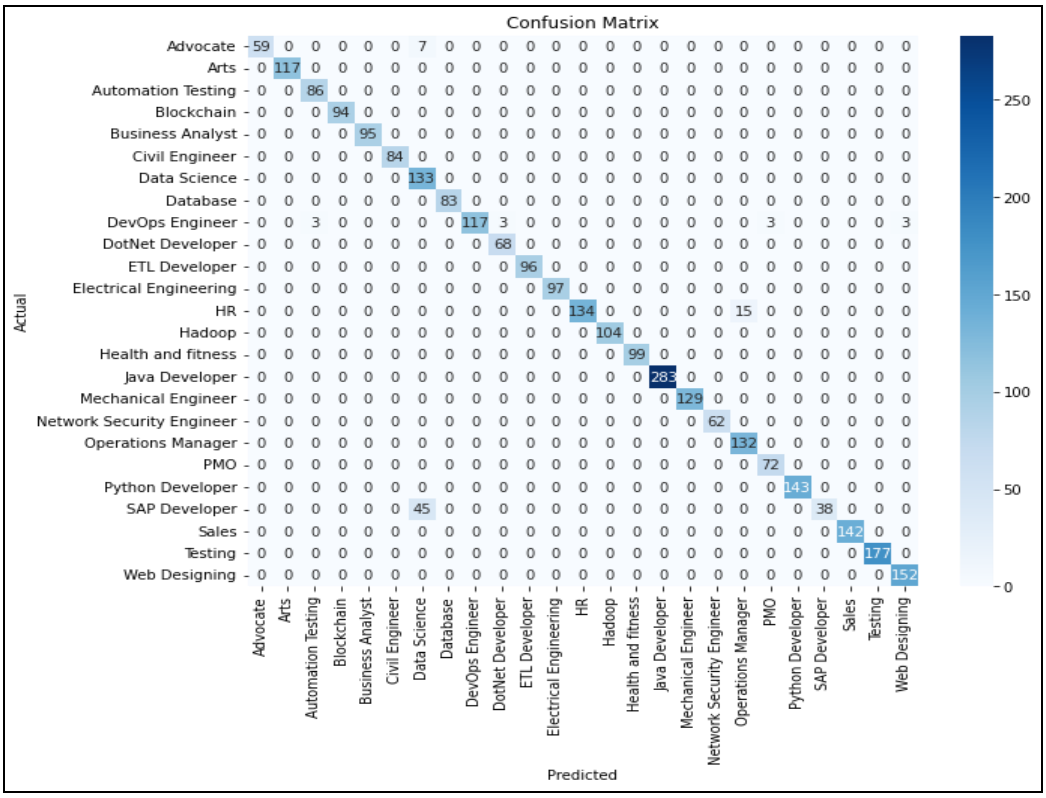

Fig. 5. Confusion matrix of the KNN algorithm

B. *Obtained Results for KNN*

After training the KNN classifier on the processed data, the total accuracy of 97.2%, precision of 97.8%, recall of 97.0% and f1-score of 97.2% has been achieved. This good accuracy show that how much efficient processed dataset is for training phase.

The figure 5 represents the Confusion matrix for the above-mentioned KNN resume analysis model.

TABLE II.

Obtained Results of KNN

Parameter	Value in percentage
Accuracy	97.2
Precision	97.8
Recall	97.0
F1-Score	97.2

V. CONCLUSION

The proposed study aims to develop an effective resume analyzer and explores the power of MapReduce for processing large datasets. As the dataset size increases, we observed the changes in the processing times which shows the adaptive and scalable nature of the MapReduce framework for handling extensive amount of data. Further, integration of KNN model for NLP tasks, resulted in 97.2% accuracy in resume categorization. Our findings focus on the role of MapReduce in processing datasets in Machine learning task and increasing the accuracy and efficiency of machine learning models.

In the future there are various paths for further research in this project. This include increasing the size of dataset by add more diverse job roles and training model on those datasets. One can also try combining different other ML algorithms to make hybrid model which will produce much better results..

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